

Universidade do Porto

Faculdade de Ciências (FCUP) | Faculdade de Engenharia (FEUP)

Departamento de Ciência de Computadores (DCC)

Licenciatura em Inteligência Artificial e Ciência de Dados

Final Project Report

The Impact of Missing Values on Uncertainty-Aware LSTM Models for Clinical Time Series

An Empirical Study on ICU Mortality Prediction

Author:

Bruno Anderson Magalhães Rocha

Supervisor:

Moisés Rocha dos Santos

June 2026

AGRADECIMENTOS

Gostaria de expressar a minha mais sincera gratidão aos meus pais e familiares, meus amigos, representadas nas pessoas de Talys, Cristóvão e João e à minha amada esposa, Yasmine, pelo apoio incondicional e pelo incentivo ao longo desta jornada que tantas horas juntos nos custaram.

Agradeço também, de forma especial, aos meus colegas e colaboradores do DCC, em particular à Professora Inês Dutra e ao Professor Francesco Renna, pela dedicação, empatia e conhecimentos recebidos que foram fundamentais para o desenvolvimento deste trabalho.

Por fim, agradeço ao meu orientador, Professor Moisés Rocha dos Santos, pelo apoio e orientação ao longo deste projeto.

Contents

1	Introduction	3
2	Framing and State of the Art	4
3	Project Description	5
3.1	Methodology	5
3.1.1	Computational Environment	5
3.1.2	Dataset	5
3.1.3	LSTM Architecture	6
3.1.4	Monte Carlo Dropout for Uncertainty Quantification	7
3.1.5	Dataset Construction: Feature Selection and Temporal Aggregation	7
3.1.6	Missing Data Handling	8
3.1.7	Train/Test Partition	8
3.1.8	Cross-Validation Design and Model Initialization	8
3.1.9	Hyperparameters and Training Protocol	9
3.1.10	Robustness Evaluation Design: Artificial Missingness Injection	9
3.1.11	Primary and Secondary Performance Metrics	10
4	Results Analysis	11
4.1	Training	11
4.2	Evaluation in the Test Group	13
4.3	Analysis of the 95% Credibility Intervals (CrI) Obtained from the MC-Dropout LSTM Predictions	15
5	Conclusion and Future Work	18
	Appendices	20
A	Bayesian LSTM via Variational Inference	21
B	Additional Discrimination and Calibration Metrics Across Missingness Scenarios	23
C	Missing-Data Statistics Across the Nine Test Scenarios	32
D	Patient Counts by Error-Agreement Category	34
	References	35

1 Introduction

Long Short-Term Memory (LSTM) networks implicitly expect a regularly sampled, fully observed input at every time step; almost every clinically derived time series, however, violates this assumption. Vital signs and laboratory results are measured at irregular intervals dictated by clinical need, and the resulting missingness is rarely independent of the outcome being predicted – patients who are more closely monitored are, by that very fact, often more severely ill [1, 2]. This *informative missingness* means that naively imputing or masking missing values can discard predictive signal or introduce bias, and conventional deterministic LSTMs offer no native mechanism to express how confident a prediction is given a particular missingness pattern. This issue is amplified in Real-World Data (RWD) from hospital information systems, which inherits the operational and clinical idiosyncrasies of the care process – equipment downtime, shift changes, selective ordering of laboratory tests – generating missingness that is simultaneously high in volume and non-random in mechanism [3]. Building predictive pipelines that are explicitly tested against, rather than merely tolerant of, these regimes is therefore a prerequisite for any model that aspires to inform an actual clinical workflow.

Mortality prediction models occupy a particularly sensitive position in this landscape: when properly validated, they function as quantitative decision-support tools that complement clinical judgement, comparable against established severity scores such as SAPS or SOFA [4]. Their practical value, however, is conditioned on whether the model’s confidence remains trustworthy precisely in the scenarios – incomplete monitoring, missing laboratory draws – that are most likely for the sickest patients. The present project sits at the intersection of applied and theoretical data science: it requires the full engineering pipeline of a real predictive system (data ingestion, feature engineering, missing-data handling, training under hardware constraints, robustness evaluation), while drawing on Bayesian approximation theory to justify Monte Carlo Dropout as an uncertainty-quantification mechanism, on calibration theory to interpret the resulting probabilities, and on statistical hypothesis testing to compare architectures and scenarios.

The overall study design – training a recurrent model on a real, naturally incomplete clinical time series and then systematically degrading it further under controlled missingness mechanisms – closely mirrors the methodology popularized by Che et al. [2] in their GRU-D study (*Recurrent Neural Networks for Multivariate Time Series with Missing Values*), and by Lipton et al. [1], who compared zero-filling, mean-imputation and missingness-indicator strategies on the same family of ICU datasets. It is also aligned with the benchmarking philosophy of Harutyunyan et al. [5], who established reproducible LSTM baselines for mortality, length-of-stay, decompensation and phenotyping tasks on MIMIC-III; while that work does not itself inject artificial missingness, its emphasis on transparent, multi-architecture comparison on a shared cohort directly informs the experimental protocol adopted here.

The full implementation underlying this report – data preparation, model training, and the robustness and uncertainty evaluation discussed below – is openly available as a code repository¹, organized as three Jupyter notebooks: `dsPrep.ipynb` consolidates the raw PhysioNet 2012 patient files and performs the feature selection, missing-data analysis and train/test split described in Section 3.1; `dtLstm.ipynb` implements and trains the Standard, Deep Ensemble and Bayesian (variational) LSTM architectures under the repeated cross-validation scheme; and `results.ipynb` injects the artificial missingness scenarios into the held-out test set and reproduces the performance and credible-interval analysis of Section 4.

¹<https://github.com/brunoamrocha/lstmBayesProject>

Primary Objective. The primary objective of this work is to analyze how a standard LSTM-based mortality classifier degrades under increasing and qualitatively different missing-data regimes, and to assess whether Monte Carlo (MC) Dropout can provide an instance-level uncertainty signal that tracks that degradation. MC Dropout, introduced by Gal and Ghahramani [7], exploits the formal equivalence between a neural network trained with dropout and an approximate variational-inference scheme over a deep Gaussian process: by leaving the dropout layers active at inference time and performing T stochastic forward passes for the same input, the resulting predictions approximate samples from the network’s posterior predictive distribution, and their dispersion can be interpreted as an estimate of epistemic uncertainty. This requires no architectural changes and reuses the very same dropout layers already present for regularization, allowing a direct comparison between the deterministic and the MC-Dropout-based behaviour of identical trained weights. As a complementary perspective, this project also implements Deep Ensembles [8], a non-Bayesian alternative that derives an uncertainty estimate from the disagreement across several independently trained networks rather than from repeated stochastic passes through one. Both strategies are evaluated under the same missing-data scenarios, so the project’s primary analysis is, more precisely, a comparison of how missingness-induced degradation is captured by these two complementary uncertainty-estimation strategies.

Secondary Objective. As a direct application of the primary analysis, the secondary objective is to build a complete in-hospital mortality prediction pipeline for ICU patients and subject it to an explicit robustness evaluation, rather than reporting a single performance figure on a single, idealized test set. Mortality-risk models of this kind are increasingly proposed as tools to support resource-allocation decisions – prioritizing ICU bed assignment, anticipating closer monitoring, or flagging patients for earlier clinical review [9] – use cases that place a premium on a model’s behaviour exactly where data are most often incomplete. The robustness analysis performed here, across missingness mechanisms, severities and inference strategies, is intended to characterize not just whether the model is accurate, but *under which conditions* its output can be trusted enough to inform this kind of consequential decision.

2 Framing and State of the Art

The challenge of handling missing values in recurrent clinical models has been addressed from two largely separate research lines, which this project deliberately brings together. The first treats missingness itself as a modelling problem: Lipton et al. [1] compared zero-filling, mean imputation and explicit missingness indicators as LSTM inputs on PhysioNet/MIMIC-III data, showing that the *pattern* of missingness carries predictive information, and Che et al. [2] formalized this into the GRU-D architecture, which incorporates a learned decay term and a masking vector directly into the recurrent unit. Both established that recurrent clinical models should be evaluated *as a function of* their missingness regime rather than under a single, fixed level of completeness – precisely the logic adopted here through the artificial-missingness injection protocol of Section 3.1. The second line treats reproducible benchmarking as the central problem: Harutyunyan et al. [5] proposed four canonical clinical-prediction benchmarks on MIMIC-III with strong LSTM baselines, and Thorsen-Meyer et al. [4] extended this into an explicitly decision-support-oriented setting, pairing an hourly LSTM for 90-day ICU mortality with model explainability to align its behaviour with clinically familiar scoring practice.

A third, more recent line concerns uncertainty quantification (UQ) and its transfer to clinical decision-making. Gal and Ghahramani [7] reframed dropout as approximate Bayesian inference

(MC Dropout), while Lakshminarayanan et al. [8] proposed Deep Ensembles as a simpler, often-competitive non-Bayesian alternative. Closer to the present design, a related benchmarking study [10] directly compared MC Dropout, Deep Ensembles and fully Bayesian recurrent networks on a MIMIC-III-style mortality task, reporting that commonly used stochastic neural networks can be poorly calibrated for clinical decision support unless carefully validated – a cautionary result that motivated training a variational (Bayesian) LSTM as a candidate third architecture here, ultimately reported separately (Appendix A) once its behaviour proved uncompetitive with the two retained architectures.

The specific gap this project addresses is that these lines are rarely combined explicitly: missingness-robustness studies in the style of [2, 1] typically report a single deterministic prediction per missingness level without an accompanying uncertainty estimate, while UQ studies in the style of [7, 8, 10] typically evaluate uncertainty under a single, fixed completeness regime. The present work crosses both axes on the same cohort and outcome – graded, mechanism-aware missingness injection (MCAR and temporal-block “monitor failure” patterns, at multiple severities) evaluated jointly with deterministic, MC-Dropout and Deep-Ensemble inference – to characterize not only *how much* performance degrades under missingness, but *how well the model knows it*. Finally, the choice of Precision-Recall AUC as the primary evaluation metric (Section 3.1.11) is grounded in the established methodological literature on evaluating binary classifiers under class imbalance [14, 15] – the regime that characterizes essentially all in-hospital mortality datasets, including the one used here.

3 Project Description

The project is organized as a three-stage pipeline, mirroring its underlying code structure: (i) data analysis and dataset preparation, which consolidates the raw PhysioNet Challenge 2012 records into a single wide-format dataset suitable for sequence modelling; (ii) model training and selection, in which three candidate LSTM-based architectures are trained under a repeated stratified cross-validation scheme; and (iii) performance evaluation on a held-out test set, in which the selected architectures are systematically re-evaluated under nine controlled missing-data scenarios (one baseline plus eight artificially induced missingness conditions), combining discrimination metrics, imbalance-aware calibration metrics, and instance-level uncertainty analysis. Sections 3.1 details the methodological choices underlying each of these three stages.

3.1 Methodology

3.1.1 Computational Environment

All experiments were implemented in **Python 3.13.7** using **PyTorch 2.9.0** as the deep learning framework. Models were trained with hardware acceleration via Apple’s **Metal Performance Shaders (MPS)** backend; the code transparently falls back to a CUDA GPU or, in its absence, to CPU execution, so that the same training pipeline can be reproduced regardless of the available hardware.

3.1.2 Dataset

The **PhysioNet/Computing in Cardiology Challenge 2012** dataset [11] was selected for three reasons: it is openly accessible without the data-use agreements or institutional credentialing required by comparable critical-care resources (e.g., MIMIC-III, eICU); it is a well-documented

clinical time series with substantial, clinically realistic missingness – on the order of 80% or more across its 36 variables even after hourly aggregation [12], precisely the property a missing-data robustness study requires; and, at $\approx 12,000$ ICU stays across three raw subsets (A, B, C), it is large enough to support a repeated cross-validation design while remaining tractable to train, on a single machine, within the time scope of the project.

3.1.3 LSTM Architecture

The classifier is a single `torch.nn.LSTM` module (2 stacked recurrent layers, hidden size 128, input dimension 7), consuming the windowed multivariate ICU time series (12 hourly time steps) prepared as described later in this section. Following PyTorch’s native recurrent-dropout mechanism, an internal dropout of $p = 0.2$ is applied between the two stacked layers; only the hidden state at the final time step is retained, passed through an additional dropout layer, and projected by a final linear layer to a single logit, converted to a mortality probability via a sigmoid activation. This same module definition is reused, unchanged, across all three inference-time strategies compared in this project: a standard deterministic forward pass; the MC-Dropout regime of Section 3.1.4, which re-activates the dropout layers at inference time; and the Deep Ensemble, which trains five independently initialized copies of this architecture per fold and averages their outputs.

Internally, each recurrent layer follows the standard gated memory-cell formulation of Hochreiter and Schmidhuber [16] (notation as in Zhang et al. [13], Chapter 10.1): an input gate I_t , forget gate F_t and output gate O_t – each a sigmoid-activated linear combination of the current input and the previous hidden state – regulate how a tanh-activated candidate state $\tilde{C}_t$ updates the cell state,

$$C_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t, \quad H_t = O_t \odot \tanh(C_t), \quad (1)$$

with $\odot$ denoting the Hadamard (element-wise) product and H_t the hidden state passed to the next layer or time step. Figure 1 depicts the resulting information flow through the cell, redrawn after the exposition in Zhang et al. [13].

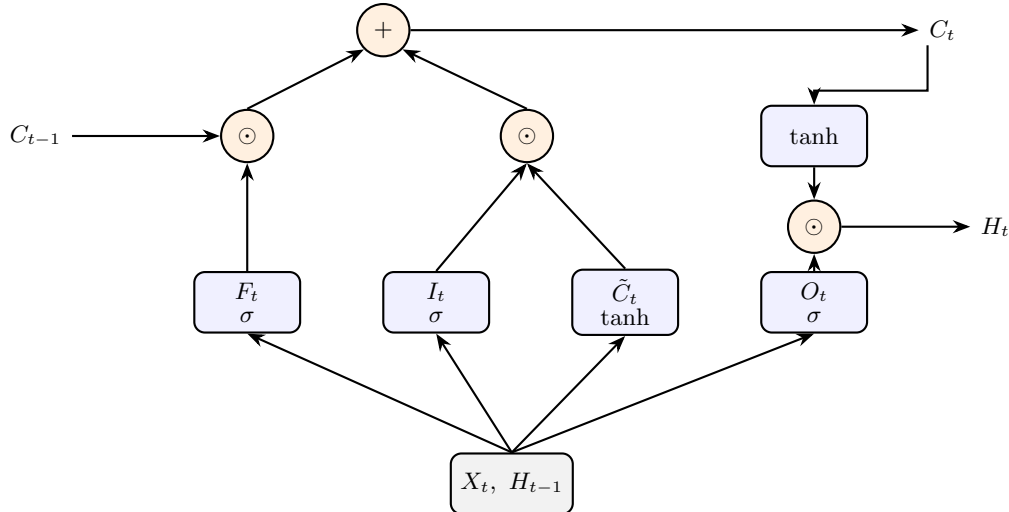


Figure 1: Schematic representation of the gated memory cell underlying the LSTM classifier used in this project, redrawn after the exposition in Zhang et al. [13] (Chapter 10.1): the input (I_t), forget (F_t) and output (O_t) gates and the candidate state $\tilde{C}_t$ are computed from X_t and H_{t-1} ; the cell state C_t and hidden state H_t are then obtained via the Hadamard product $\odot$ as in Eq. 1.

3.1.4 Monte Carlo Dropout for Uncertainty Quantification

The classification head of the network applies dropout with probability $p = 0.2$ both between the two stacked LSTM layers (PyTorch’s native recurrent dropout) and immediately before the final fully connected (linear + sigmoid) classification layer. Under standard practice, all dropout layers are disabled at inference time (`model.eval()`). For the MC-Dropout configuration, this behaviour is deliberately overridden: every module whose class is a dropout layer, together with the recurrent LSTM module itself, is forced back into training-mode behaviour during inference, while the rest of the network (e.g., batch statistics, if any) remains in evaluation mode. As formalized by Gal and Ghahramani [7], repeating the forward pass under this regime is mathematically equivalent to drawing samples from an approximate variational posterior over the network’s weights. Concretely, for every patient instance in the evaluation set, the network performs $T = 50$ independent stochastic forward passes; the point prediction is taken as the mean of these 50 Monte Carlo samples, and the 2.5th and 97.5th percentiles of the resulting empirical distribution define an approximate 95% credible interval, which is used downstream as the patient-level uncertainty estimate. The same 50-sample procedure is also used, equivalently, during model *validation* for the Bayesian (variational) architecture reported in Appendix A, where the 50 samples are averaged to estimate the predictive mean prior to computing the validation loss.

3.1.5 Dataset Construction: Feature Selection and Temporal Aggregation

The raw PhysioNet 2012 records (one file per patient, in long format, with irregular timestamps over the first 48 hours of ICU stay) were first consolidated across the three raw subsets and merged with the corresponding outcome file (in-hospital death label, SAPS-I, SOFA, length of stay and survival time). From the 37 available time-series variables, a candidate subset of 12 clinically relevant parameters with high observation density (Weight, GCS, HR, NIDiasABP, NIMAP, NISysABP, RespRate, Temp, DiasABP, MAP, SysABP, Urine) was retained and aggregated into fixed, non-overlapping one-hour bins (taking the mean when multiple readings fell within the same hour), restricted to the first **12 hours** of ICU stay (t_0 to t_{11}). This restriction to a 12-hour, hourly-resolution window was chosen to keep the resulting sequence length compatible with the scope and the computational budget of the project, while still covering the period in which early clinical deterioration is typically detectable.

Within this 12-parameter candidate set, the three pairs of invasive and non-invasive blood-pressure measurements (DiasABP/NIDiasABP, MAP/NIMAP, SysABP/NISysABP) were merged into three single features (**DiasABPf**, **MAPf**, **SysABPf**) by prioritizing the invasive reading and falling back to its non-invasive counterpart whenever the former was missing. Two variables (**RespRate** and **Weight**) were subsequently dropped due to excessive missingness, and patients with insufficient valid observations for any remaining feature (fewer than 3 valid hourly readings out of 12 for GCS, Urine and Temp; fewer than 6 out of 12 for the remaining variables) were excluded from the cohort. This procedure yielded the final set of **seven medical parameters** used to train every LSTM in this project: **GCS**, **Heart Rate (HR)**, **Temperature (Temp)**, **Urine output**, and the three fused blood-pressure channels (**DiasABPf**, **MAPf**, **SysABPf**). The resulting table was pivoted into a **wide-format** representation, with one row per patient and $7 \text{ features} \times 12 \text{ time steps} = 84$ feature columns, in addition to the patient identifier and the binary outcome label.

After this filtering process, the final cohort consisted of **8,416 patients**, with a strongly imbalanced outcome distribution: 7,189 survivors (84.5%) and 1,227 in-hospital deaths (15.5%). It is worth noting that, even within this curated subset, baseline missingness remained substantial and heterogeneous across features: GCS, Urine and Temp started with as few as 3 valid observations

out of 12 (i.e., 75% missing) for some patients, while the remaining four parameters started with around 6 valid observations out of 12 (50% missing) – underscoring, once more, the suitability of this dataset for the missingness-robustness analysis that motivates the project.

3.1.6 Missing Data Handling

Missing values were imputed using a deterministic, two-tier strategy applied independently to each of the seven features, within each patient’s own 12-hour timeline:

1. **Forward fill (LOCF), horizontally.** Within a patient’s timeline, a missing value at time t is replaced by the most recent previously observed value for that same feature, under the assumption that the most recent clinical state is the best available predictor of the current one.
2. **Row-mean imputation.** If the very first observation(s) of a patient’s timeline are missing – a case forward-fill cannot resolve, since there is no earlier value to carry forward – the remaining gaps are filled with the arithmetic mean of that patient’s own available measurements for that feature.

This two-tier, patient-specific imputation was applied identically to the training data and to every artificially degraded version of the test data (Section 3.1.10), ensuring that any change in downstream performance can be attributed to the missingness itself rather than to inconsistencies in the imputation procedure.

3.1.7 Train/Test Partition

Before any model development took place, the full cohort was split into an 80% training set (6,732 patients) and a 20% held-out test set (1,687 patients), stratified jointly on (i) the in-hospital death label and (ii) a derived missingness-richness indicator for the three features with the highest baseline missingness (GCS, Urine, Temp), each categorized as *Low* (fewer than 6 valid observations out of 12) or *Normal* (6 or more). Rare combinations of these strata were merged into the most frequent stratum to avoid split failures. This design ensures that the train and test partitions are balanced not only with respect to mortality prevalence, but also with respect to the underlying missingness profile of the cohort. The resulting partitions preserved the original class ratio almost exactly: 85.4% survivors / 14.6% deaths in both the training set (5,751 vs. 981 patients) and the test set (1,468 vs. 246 patients). The 20% test set was set aside exclusively for the final evaluation described in Section 3.1.10 and was not used, in any form, during model training or selection.

3.1.8 Cross-Validation Design and Model Initialization

Model selection was performed exclusively on the 80% training partition, using a **Repeated Stratified 5-fold Cross-Validation**, repeated 3 times (**Repeated Stratified K-Fold**, $5 \times 3 = 15$ folds in total, fixed random seed), so that every fold preserves the original $\approx 85/15$ class ratio. No oversampling or undersampling strategy was applied at any stage; class imbalance was instead handled exclusively through the choice of imbalance-aware evaluation metrics (Section 3.1.11), so as not to distort the native missingness and imbalance structure of the data that the project specifically aims to study.

For each of the 15 folds, every architecture was trained completely from scratch: weights were freshly initialized and mini-batch shuffling was independently re-sampled, so that the 15 trained instances of a given architecture are not 15 evaluations of a single fixed solution, but 15 independent

draws from the LSTM’s stochastic training process. This design serves two purposes. First, it allows the natural training variability of each architecture to be characterized explicitly (e.g., by reporting best-, median- and worst-performing folds, rather than a single run). Second, it directly supports the Deep Ensemble architecture, for which 5 independently initialized and trained members were produced *per fold* ($5 \times 15 = 75$ networks in total for this architecture), with their predictions averaged at inference time. The standard (deterministic) architecture’s 15 fold-trained networks are, in turn, reused twice at inference time: once under a standard deterministic forward pass, and once under the MC-Dropout regime described in Section 3.1.4 – meaning that “standard” and “standard (MC Dropout)” correspond to the exact same trained weights, evaluated under two different inference-time procedures, rather than to two independently trained models.

A third candidate architecture – a Bayesian LSTM trained via variational inference – was also implemented and trained under the same cross-validation scheme. Given the training and calibration behaviour observed for this variant, it was excluded from the comparative analysis that follows; its implementation, training setup and results are reported separately in Appendix A.

3.1.9 Hyperparameters and Training Protocol

All retained architectures (standard LSTM and Deep Ensemble) share an identical set of hyperparameters, summarized in Table 1, so that differences in performance can be attributed to the architecture and to the missing-data scenario rather than to inconsistent tuning.

Table 1: Shared hyperparameters and training protocol.

Hyperparameter	Value
Recurrent layers (stacked)	2
Hidden dimension	128
Dropout probability	0.2
Output layer	1 unit, sigmoid (binary classification)
Optimizer	Adam, initial learning rate = 0.001
Learning-rate schedule	Adaptive (Reduce On Plateau on validation loss)
Loss function	Binary Cross-Entropy (BCE)
Batch size	64
Maximum epochs	50
Early stopping(optional)	Patience = 15 epochs, $\min \Delta = 10^{-5}$
Ensemble size (Deep Ensemble)	5 members per fold
MC-Dropout samples (test-time)	50 stochastic forward passes per instance
Cross-validation scheme	5×3 Repeated Stratified K -Fold (15 folds)

3.1.10 Robustness Evaluation Design: Artificial Missingness Injection

To assess robustness beyond the dataset’s native missingness, the held-out test set was further degraded under nine controlled scenarios: a 0% baseline (the test set as constructed, before any additional missingness) and eight artificially injected conditions, combining two missingness *mechanisms* with four injected *severities*, as summarized in Table 2.

Table 2: Missing-data injection scenarios applied to the test set.

Mechanism	Description	Severities
MCAR	Missing Completely at Random: valid observations are removed uniformly at random within each patient’s timeline for a given feature.	10%, 20%, 40%, 60%
Temporal Block	Simulates a contiguous monitor failure: a random failure onset is chosen and the nearest available observations (in time) are progressively removed.	10%, 20%, 40%, 60%

In both mechanisms, the injected percentage is computed relative to the number of *currently available* (non-missing) observations for that patient and feature, guaranteeing at least one removed value whenever at least one valid observation exists. After injection, the same two-tier imputation strategy described in Section 3.1.6 was reapplied, so that every scenario is evaluated through an identical downstream pipeline and only the missingness mechanism and severity vary across conditions.

3.1.11 Primary and Secondary Performance Metrics

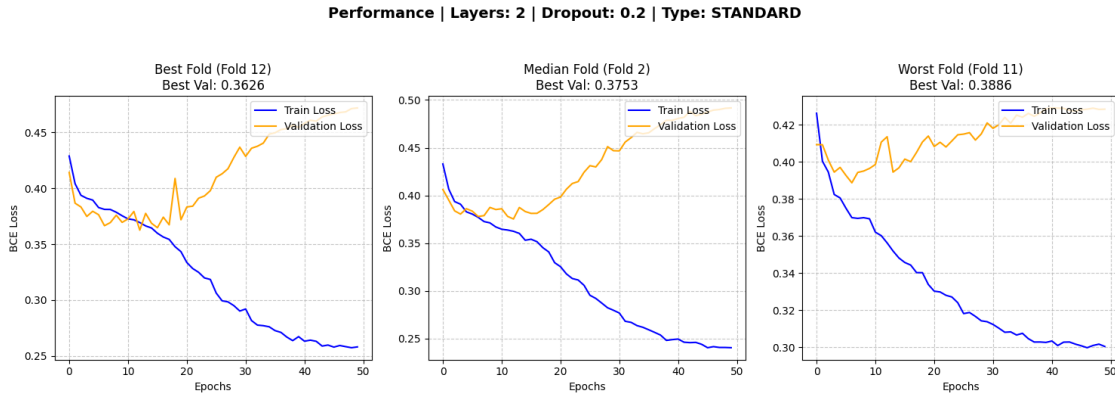
Given the substantial class imbalance of the outcome ($\approx 14\text{--}15\%$ positive rate, consistently across training, test and all injected scenarios), the **area under the Precision-Recall curve (AUC-PR)** was adopted as the *primary* evaluation metric throughout this project. This choice follows directly from the methodological literature on evaluating binary classifiers under class imbalance. Saito and Rehmsmeier [14] showed that ROC curves can be visually and numerically misleading on imbalanced datasets, because the false positive rate (the ROC x -axis) is computed over the majority class and therefore tends to remain favourable even when a classifier generates a large absolute number of false positives relative to the (small) positive class; Precision, by construction, is sensitive to exactly this failure mode, since it is computed over the set of *positive predictions* rather than over the negative class. Davis and Goadrich [15] further established the formal relationship between ROC and PR space, showing that a curve dominating in ROC space also dominates in PR space, but that algorithms optimized for AUC-ROC are *not* guaranteed to optimize AUC-PR – making the explicit choice between the two metrics a methodologically consequential one rather than a matter of convention. Like AUC-ROC, AUC-PR is threshold-independent, summarizing classifier performance across the full range of decision thresholds; this property is particularly relevant here, since different architectures and different missingness scenarios are expected to shift the operating point of the classifier, and a threshold-independent metric is required for a fair comparison across them.

AUC-ROC was nonetheless also computed and reported as a *secondary*, complementary diagnostic during the model-selection (training and cross-validation) stage, for two reasons: first, to remain comparable with the substantial body of prior LSTM-based clinical time-series literature that conventionally reports AUC-ROC [4, 5]; and second, to allow detection of the specific dissociation pattern in which a model’s discrimination appears to hold under AUC-ROC while collapsing under AUC-PR – a known signature of imbalance-sensitive overestimation of performance when ROC-based metrics are used in isolation.

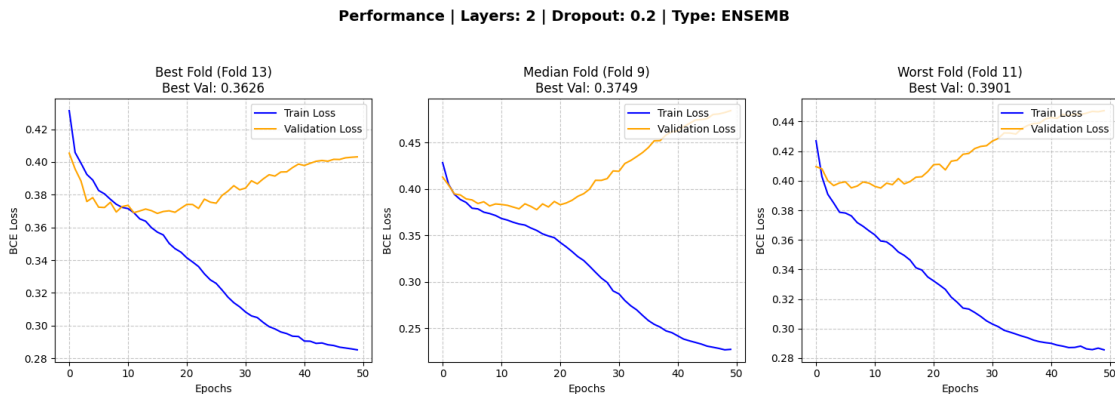
4 Results Analysis

4.1 Training

Figure 2 presents the training dynamics of the two retained architectures across the 5×3 repeated stratified cross-validation described in Section 3.1.8. For each architecture, the three folds with the best, median and worst final validation loss (out of the 15 trained instances) are shown, plotting training loss against validation loss over the 50 training epochs. A consistent pattern emerges across both architectures and across all three representative folds: training loss decreases monotonically throughout, but validation loss only tracks it down to approximately **epoch 20**, after which it begins to climb again while training loss continues to fall – the classic signature of the network increasingly fitting fold-specific noise in the training partition rather than generalizing. Despite this divergence, the best validation loss reached before early stopping intervenes is remarkably consistent across folds: 0.363 (best fold), 0.375 (median fold) and 0.389 (worst fold) for the Standard LSTM (fig. 2a), and 0.363, 0.375 and 0.390, respectively, for the Deep Ensemble (fig. 2b) – a spread of only about 0.026 BCE units between the best- and worst-performing fold for either architecture, and an essentially identical range between the two architectures themselves.



(a) Standard LSTM



(b) Deep Ensemble (5-member average per fold)

Figure 2: Training (blue) and validation (orange) BCE loss over 50 epochs, for the best, median and worst performing cross-validation fold (out of 15), (a) for the Standard LSTM and (b) Deep Ensemble.

The same best/median/worst folds are re-examined in Figure 3 through their discrimination curves on the validation partition. Here, the contrast between folds is considerably **more visible** than in the loss curves: for the Standard LSTM, AUC-ROC ranges from 0.755 (best fold) to 0.683 (worst fold), a spread of 0.072, while AUC-PR ranges from 0.364 to 0.267, a spread of 0.097 – proportionally a much larger relative difference on the PR scale ($\approx 30\%$ of the best value) than on the ROC scale ($\approx 10\%$), consistent with PR curves being more sensitive to class-imbalance-driven instability across folds. The same pattern holds for the Deep Ensemble (AUC-ROC: 0.761/0.731/0.694; AUC-PR: 0.369/0.343/0.276 for best/median/worst). Critically, when the two architectures are compared fold-by-fold rather than within themselves, the Standard LSTM and the Deep Ensemble are remarkably **similar** on both metrics (within 0.006 AUC-ROC and 0.012 AUC-PR at matching quantiles) – the cross-fold variability of a single architecture is larger than the architecture-to-architecture difference at any given quantile of performance.

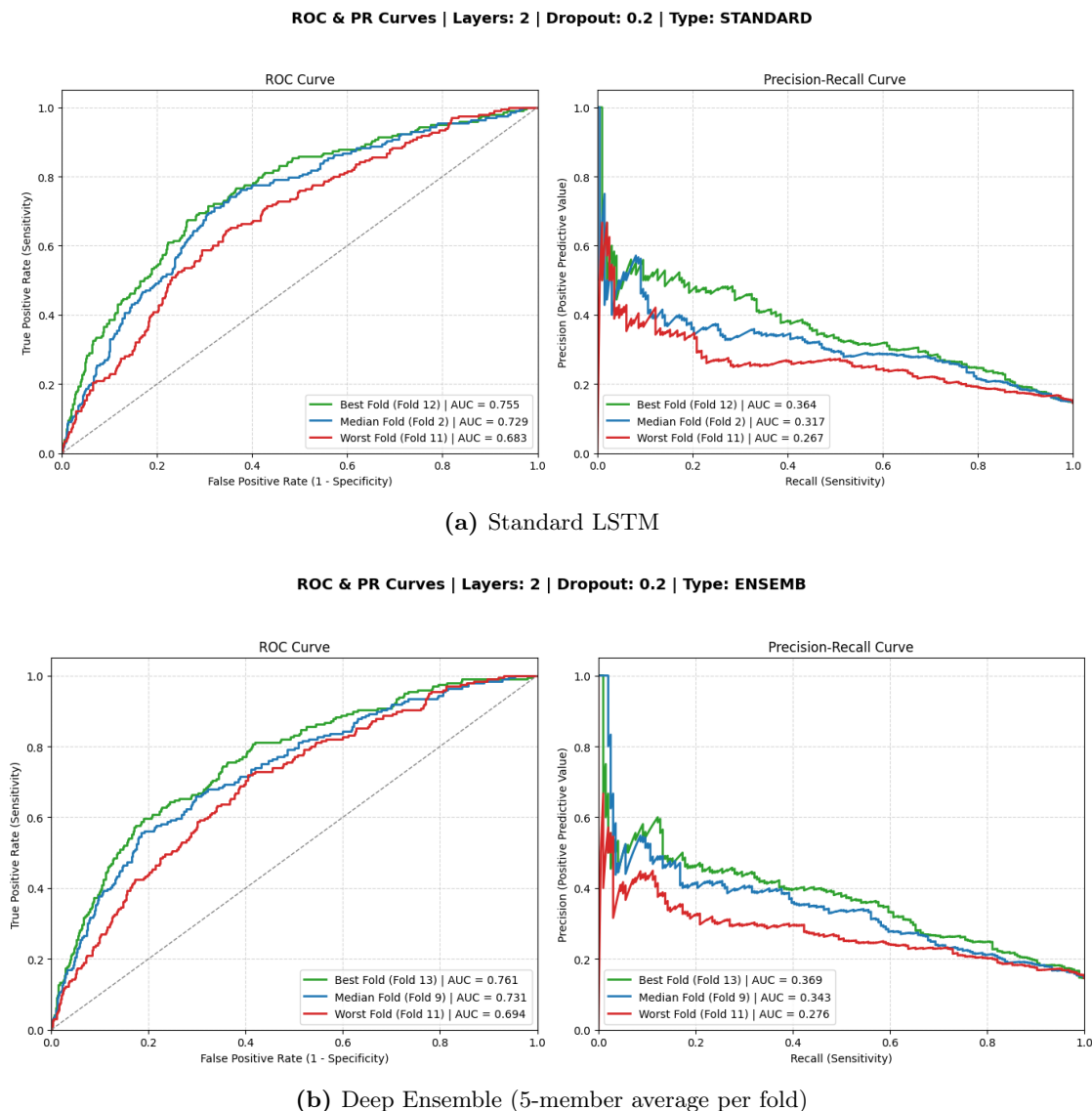


Figure 3: ROC (left) and Precision-Recall (right) curves on the validation partition for the best, median and worst cross-validation fold, for the Standard LSTM (top) and the Deep Ensemble (bottom). The best/median/worst spread is visibly larger on the PR curves than on the ROC curves.

4.2 Evaluation in the Test Group

Section 3.1.10 described how the held-out test set was further degraded under nine controlled missingness scenarios (a 0% baseline plus MCAR and Temporal-Block injection at 10%, 20%, 40% and 60%). One detail of that procedure is worth restating here, since it directly shapes the results below: the "one at least" rule guarantees that, whenever the missing data rate to be introduced is less than one, *at least one* value is always removed, even when the nominal injected percentage would otherwise round down to zero. Combined with the substantial *native* missingness already present in the test cohort before any injection: 26.5% on average across the seven features at the 0% baseline, rising to 34.8%, 40.8%, 55.8% and 70.7% on average at the 10%, 20%, 40% and 60% injection levels, respectively (Table 4, Appendix C) – this means that even the lowest injected severity already represents a non-trivial perturbation. GCS and Temperature in particular already exceed 50% missingness at baseline (60.7% and 56.4%, respectively), so the *additional* gap between, say, the 10% and 20% levels is comparatively small relative to the gap between 0% and 10% – an effect that recurs throughout the analysis below.

For every scenario, three inference strategies were applied to the same 15 fold-trained networks: the Standard LSTM evaluated deterministically; the same Standard LSTM evaluated under the MC-Dropout procedure of Section 3.1.4 (50 stochastic forward passes, averaged); and the Deep Ensemble (5 members, averaged). This produced the $3 \times 9 \times 15 = 405$ -row evaluation table (`models_metrics.csv`) underlying the analysis that follows.

Figure 4 presents, for every scenario, the mean AUC-PR ($\pm 95\%$ CI) of the three inference strategies, together with the corresponding omnibus (repeated-measures ANOVA and Friedman) and Holm-corrected pairwise Wilcoxon tests; the equivalent plots and summary tables for AUC-ROC, Balanced Accuracy, the class-balanced Brier score and the Expected Calibration Error are reported in Appendix B for completeness. Three observations recur across every scenario. **First**, the Standard LSTM and its MC-Dropout counterpart are, as expected from Section 3.1.4, statistically indistinguishable: since they are literally the same trained weights evaluated under two different inference procedures, their mean AUC-PR differs by at most 0.0003 in any scenario, and the paired Wilcoxon test comparing them is never significant ($p \geq 0.45$ in every case, frequently $p > 0.9$). **Second**, the Deep Ensemble is consistently the best-performing strategy – by a small margin – across all nine scenarios (e.g., 0.275 vs. 0.268 at 0%; 0.235 vs. 0.212 at 60% MCAR), but this advantage only reaches statistical significance, after Holm correction, in isolated scenarios (most clearly at 10% Temporal, $p \approx 0.02$ for both pairwise comparisons against the ensemble); in most scenarios the omnibus tests themselves are non-significant, and where they are significant the pairwise post-hoc comparisons are usually not, indicating a real but modest, and not consistently detectable, advantage for the ensemble. **Third**, comparing these test-set values against the fold-level cross-validation values reported in Section 4.1, the test-set mean AUC-PR for the Standard LSTM at the same 0% baseline (0.268) is almost identical to the single *worst* cross-validation fold (0.267), and well below the median (0.317) and best (0.364) fold – consistent with a modest degree of overfitting to the training partition during model selection, despite early stopping.

Figure 5 reorganizes the same AUC-PR values into a forest plot per algorithm, with missingness severity (%NA) on the vertical axis and the two injection mechanisms shown side by side. The dominant pattern is an essentially monotonic **decline in AUC-PR as %NA increases**, for all three inference strategies (omnibus RM-ANOVA across the nine scenarios is significant at $p < 10^{-55}$ in every panel): from 0.268–0.275 at the 0% baseline down to 0.212 at 60% Temporal missingness, a relative drop of roughly 20–23%. The decline is not perfectly uniform across mechanisms, however: at every severity level, the Temporal-Block mechanism produces a lower AUC-PR than MCAR at the same nominal percentage (e.g., at 60%: 0.212 Temporal vs. 0.228–0.235 MCAR across the three

AUC-PR | Paired Statistical Analysis Grid

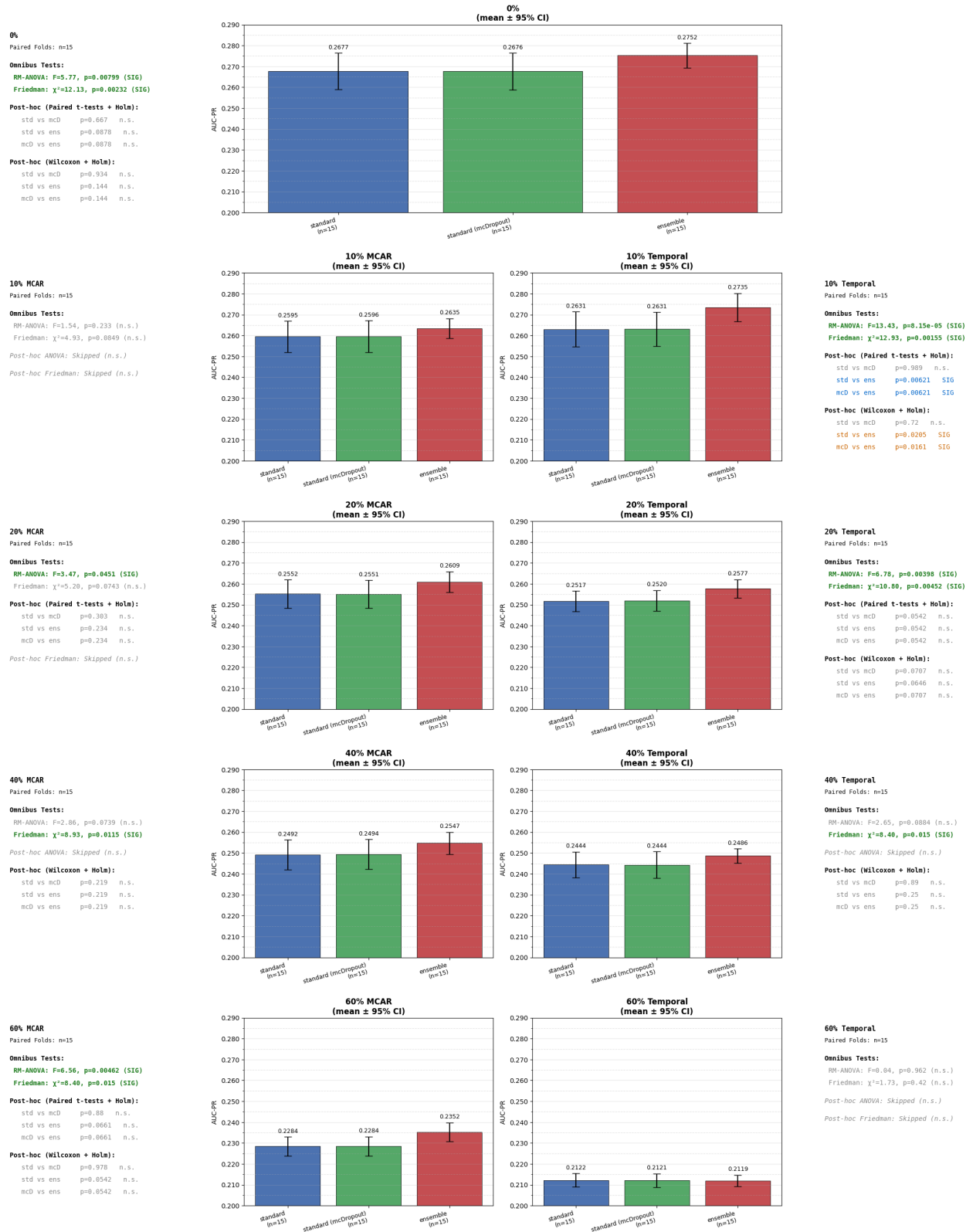


Figure 4: AUC-PR (mean $\pm$ 95% CI) for the three inference strategies (standard, MC-Dropout, ensemble; $n = 15$ folds each) across all nine missingness scenarios (left column), together with omnibus repeated-measures ANOVA and Friedman tests and, where significant, Holm-corrected pairwise Wilcoxon signed-rank post-hoc comparisons (right column).

strategies), indicating that losing a contiguous block of observations – as a real monitor failure would – is more damaging to the model than losing the same number of observations scattered at random.

4.3 Analysis of the 95% Credibility Intervals (CrI) Obtained from the MC-Dropout LSTM Predictions

To facilitate a more contextualized analysis, a categorization framework was developed based on the models’ probabilistic predictions. The predictions were classified into four distinct groups: *confident correct*, *confident error*, *borderline correct*, and *borderline error*. A decision threshold of 14.5% – matching the test set’s overall mortality prevalence – was adopted to determine correct and incorrect classifications, and a margin of approximately 20% around this threshold was applied, so that predictions between 11.5% and 17.5% were considered *borderline*, while predictions outside this range were classified as *confident*. Prediction values were taken from the three fold-level predictors – Standard fold 2, MC-Dropout fold 2, and Ensemble fold 9 – that yield the median AUC-PR (and, jointly, the median AUC-ROC) value across the K -fold cross-validation, since these metrics are threshold-independent and the median-performing fold provides a more representative measure of each algorithm’s typical behaviour, mitigating selection bias across validation folds [6]. The use of AUC-PR, in particular, is highly recommended for evaluating mortality prediction in intensive care settings due to the severe class imbalance characteristic of this task, following established clinical machine-learning benchmarking practice [5].

Because clinical decision-making around resource prioritization typically operates in terms of *confidence* rather than a single threshold value, and because the uncertainty introduced by the choice of any one threshold is itself a source of ambiguity, the subsequent analysis collapses these per-predictor labels into three coarser, more decision-relevant groups: **Confident Correct** and **Confident Error** contain only those patients for which all three predictors unanimously agreed on a confident correct (respectively, confident error) classification; every other case – borderline for at least one predictor, or disagreeing between confident correct and confident error across predictors – is pooled into a **Gray Zone**.

Figure 6 tracks the absolute number of patients in each of the four original categories (before the three-group collapse) as a function of %NA, separately for the three predictors and the two missingness mechanisms. Confident-correct counts decline and confident-error counts rise fairly consistently across all three predictors as %NA increases, as expected. This trend is noticeably less clean between the 10% and 20% levels, however, where the curves are sometimes flat or even reverse direction – consistent with the already-substantial baseline missingness established in Section 4.2, which makes the *additional* perturbation between a 10% and a 20% injection comparatively small, so a less monotonic response in this range is unsurprising. The borderline-correct counts show a similar, if less pronounced, declining trend, while the borderline-error counts behave more erratically, without a clear global trend – plausibly because this category simultaneously gains entrants from the (shrinking) confident-correct pool while losing some of its own members to the (growing) confident-error pool. Taken together with the categorical counts in Table 5 (Appendix D), these patterns are consistent with the broader expectation, supported by the missingness-handling literature [1, 2], that recurrent clinical classifiers degrade as the amount of available information per instance decreases.

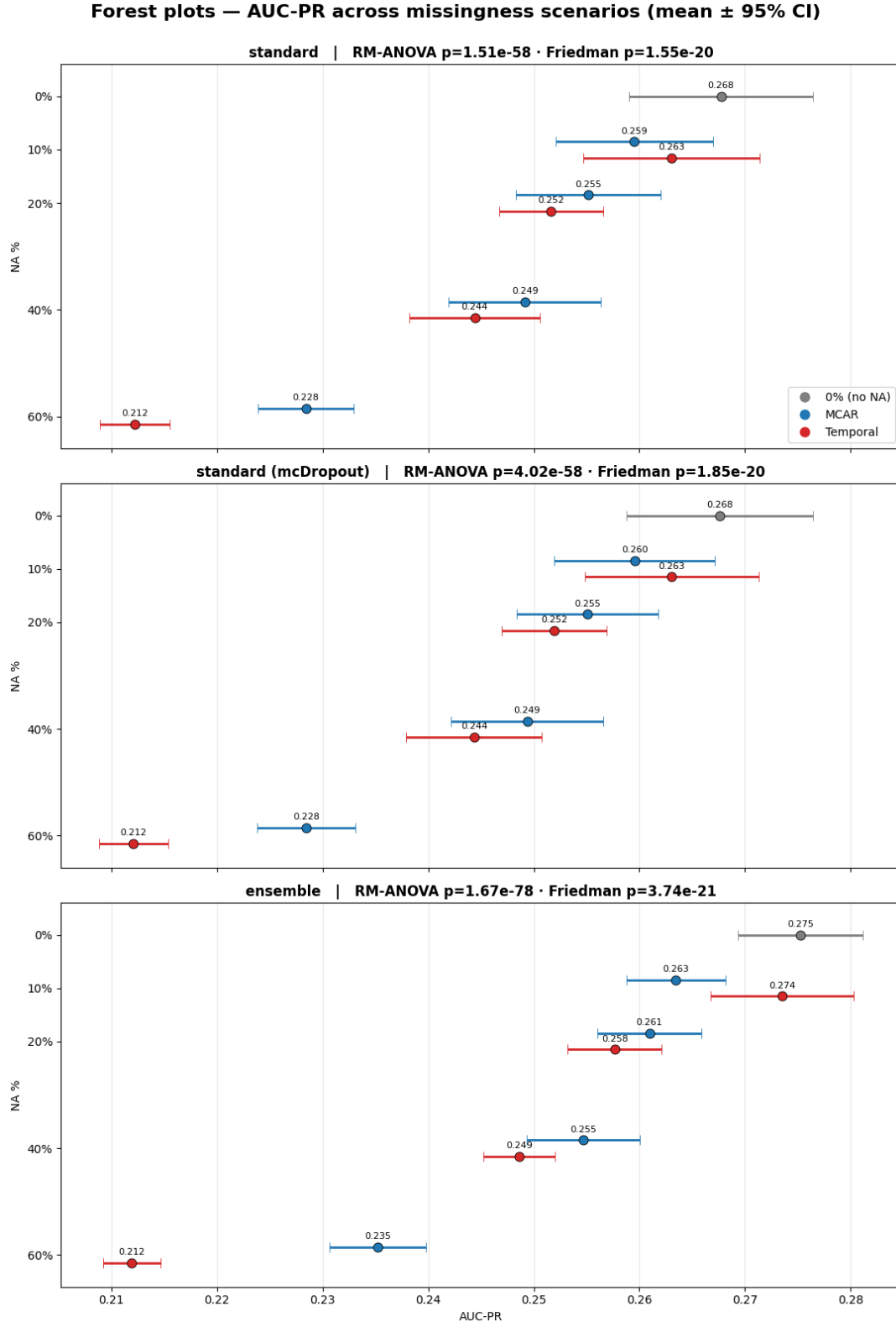


Figure 5: Forest plot of mean AUC-PR ($\pm 95\%$ CI) as a function of missingness severity (%NA), separately for the Standard LSTM (top), MC-Dropout (middle) and Deep Ensemble (bottom), and for the MCAR (blue) and Temporal-Block (red) mechanisms. The grey 0% point is the shared baseline. RM-ANOVA and Friedman p -values across the nine scenarios are annotated per panel.



Figure 6: Absolute number of patients in each error-agreement category (Confident Correct, Confident Error, Borderline Correct, Borderline Error; 2×2 grid) across missingness severity (%NA), for the Standard, MC-Dropout and Ensemble predictors (colors) under MCAR (solid lines, circles) and Temporal-Block (dotted lines, squares) missingness, all referenced to a shared 0% baseline.

Figure 7 examines the credible-interval width itself – the difference between the 97.5th and 2.5th percentiles of the 50 MC-Dropout samples per instance (Section 3.1.4) – as a function of the three-group classification just described, across all nine missingness scenarios. The pattern is consistent and, from a decision-support perspective, encouraging: the mean CrI95% width is smallest for the **Confident Correct** group (≈ 0.052 – 0.053 across all scenarios), intermediate for the **Confident Error** group (≈ 0.055 – 0.057), and largest for the **Gray Zone** (≈ 0.059 – 0.060), with this ordering holding at every missingness level and for both mechanisms. In other words, the MC-Dropout uncertainty estimate is, on average, already informative about whether a prediction is likely to be a confident success, a confident mistake, or a genuinely ambiguous case – even though the three groups’ interval widths are themselves fairly stable across %NA (the within-group range across scenarios is only about 0.001–0.002), so it is the *separation between groups*, more than any sharp widening of the interval under higher missingness, that carries the diagnostic signal here. This is in line with the theoretical interpretation of MC-Dropout’s predictive variance as an approximate posterior-predictive uncertainty [7]: instances the model is structurally unsure about (the Gray Zone) receive a systematically wider credible interval than instances it confidently and correctly classifies.

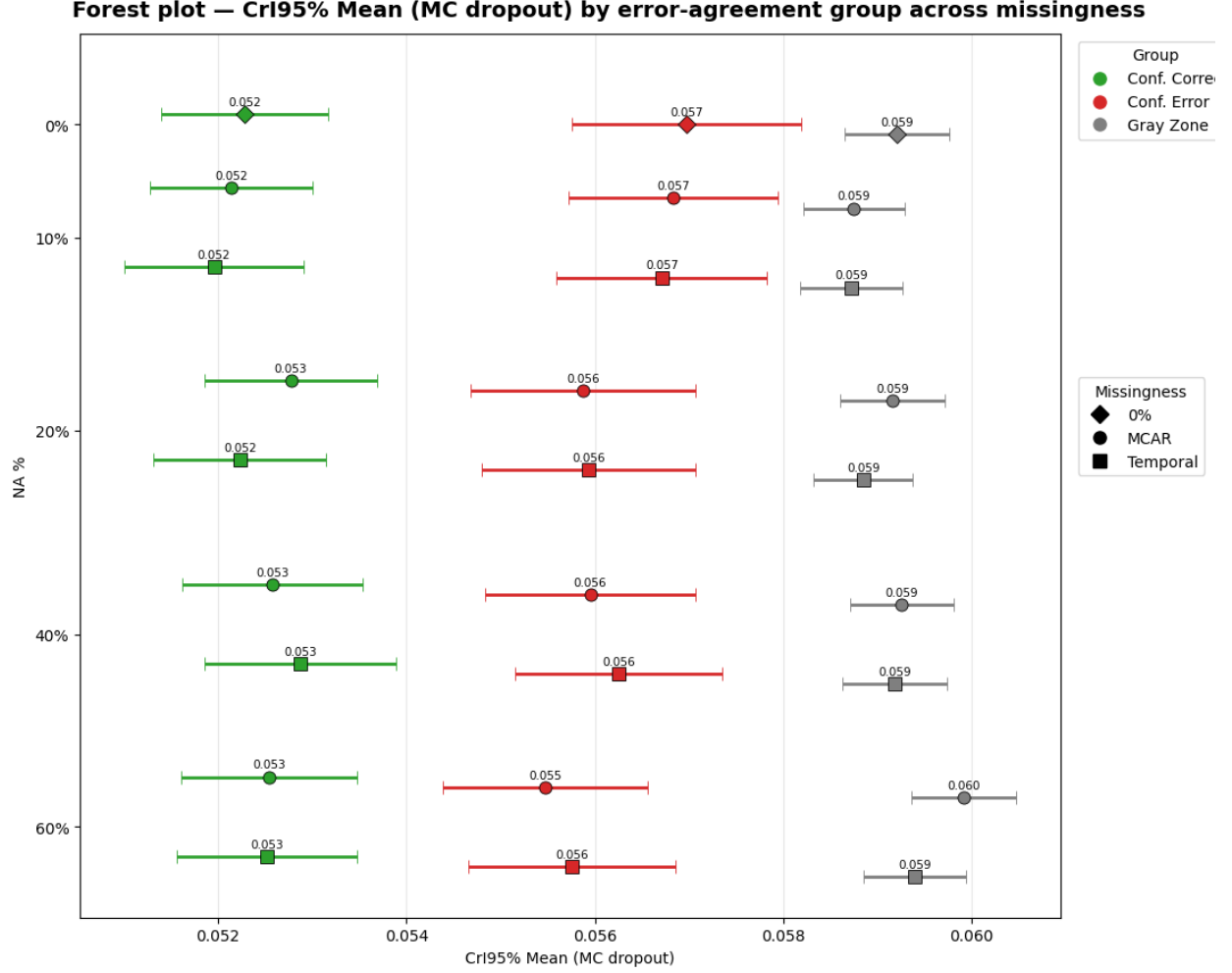


Figure 7: Forest plot of the mean 95% credible-interval width (CrI95%, MC-Dropout) by error-agreement group (Confident Correct, green; Confident Error, red; Gray Zone, grey) across missingness severity (%NA, vertical axis) and mechanism (diamond = 0%, circle = MCAR, square = Temporal). Error bars show the 95% CI of a one-way comparison (ANOVA / Kruskal-Wallis, with Holm-corrected pairwise post-hoc where the omnibus test is significant) across the three groups within each scenario.

5 Conclusion and Future Work

This project trained and systematically stress-tested two LSTM-based architectures – a Standard deterministic LSTM (additionally queried via MC Dropout at inference time) and a Deep Ensemble of five members – for in-hospital mortality prediction on the PhysioNet 2012 cohort, under nine controlled missing-data scenarios spanning two mechanisms (MCAR and Temporal-Block) and four severities. Both architectures showed similarly modest cross-validation variability during training (Section 4.1) and a consistent, statistically detectable decline in AUC-PR as missingness increased on the held-out test set (Section 4.2), with the Temporal-Block mechanism degrading performance more than MCAR at matching severity. The Deep Ensemble outperformed the Standard LSTM by a small but fairly consistent margin, though this difference rarely reached statistical significance after correction for multiple comparisons, while MC Dropout, as expected, behaved statistically indistinguishably from its deterministic counterpart in terms of point-prediction quality – its value

lies not in the point estimate itself but in the accompanying credible interval. That credible interval proved to carry real diagnostic information: its width reliably separated confidently correct predictions from confident errors and from genuinely ambiguous (Gray Zone) cases, across every missingness scenario tested (Section 4.3), supporting the secondary objective of building a mortality model whose robustness – and whose own self-assessed reliability – can be characterized explicitly rather than assumed.

Two directions seem most promising for future work. **First**, since the present comparison used a single, moderately sized cohort (8,416 patients), and dropout- or ensemble-based uncertainty estimation is reported to offer its largest relative benefit over deterministic point estimates precisely in such small-to-moderate data regimes [17], replicating this protocol on a substantially larger cohort (e.g., MIMIC-III or eICU) would clarify whether the MC-Dropout benefit observed here – and the competitiveness of the Bayesian LSTM of Appendix A [10] – persists at scale. **Second**, since CrI95% width tracked error-agreement group more reliably than missingness severity itself (Section 4.3), it could plausibly support an operational triage rule – flagging any prediction whose credible interval exceeds a calibrated threshold for mandatory clinical review – whose precision-recall trade-off, and comparison against the Deep Ensemble’s analogous disagreement signal, is a direct extension of the analysis presented here.

Appendices

A Bayesian LSTM via Variational Inference

As noted in Section 3.1.8, a third candidate architecture was trained alongside the standard LSTM and the Deep Ensemble: a fully Bayesian recurrent network in which both the recurrent layers and the final linear classification layer are replaced by their variational counterparts (`BayesianLSTM` and `BayesianLinear`, as implemented in the BLITZ library), trained end-to-end via variational inference. Each Bayesian layer maintains a learned distribution (mean and variance) over its weights rather than a single point estimate, and the network is trained by minimizing a stochastic approximation of the negative Evidence Lower Bound (ELBO), combining the standard Binary Cross-Entropy data term with a Kullback–Leibler (KL) divergence penalty between the learned weight posterior and a prior distribution, scaled by $1/N_{\text{train}}$ (where N_{train} is the number of training instances in the fold) so that the relative weight of the regularization term decreases appropriately as more data become available.

This architecture was trained under the same 5×3 repeated stratified cross-validation scheme described in Section 3.1.8, using a longer training budget of up to 100 epochs (versus 50 for the deterministic architectures), since variational training is known to converge more slowly; validation loss at each epoch was estimated by averaging 25 Monte Carlo forward passes per batch, in the same spirit as the MC-Dropout procedure described in Section 3.1.4, but applied here to a network whose stochasticity originates from sampled weights rather than from dropout masks.

Rationale for exclusion from the main analysis. While this variational architecture is, in principle, the most direct Bayesian analogue to the MC-Dropout and Deep-Ensemble approaches compared in the main text, its training and validation behaviour in this project did not reach a level of stability and calibration competitive with the two retained architectures – a difficulty consistent with prior reports that variational recurrent networks can be substantially harder to tune and can yield poorly calibrated predictive distributions in clinical decision-support settings unless carefully validated [10]. For this reason, the Bayesian LSTM was **excluded from the comparative robustness analysis** reported in the main text, and is documented here for completeness and as a basis for future refinement.

Training results. Figure 8 shows the same best/median/worst-fold diagnostic used in Section 4.1 for this architecture, plotting the training ELBO against the validation BCE on twin axes (note the different scales). Unlike the Standard LSTM and the Deep Ensemble, the Bayesian LSTM does not show a clear overfitting inflection: validation BCE oscillates considerably in the first ~ 20 epochs and then plateaus, settling at 0.4156 (best fold), 0.4160 (median fold) and 0.4172 (worst fold) – an extremely tight range across folds (0.0016), but at a validation-loss *level* clearly higher than the 0.363–0.390 range reached by the Standard LSTM and the Deep Ensemble (Section 4.1).

Performance | Layers: 2 | Dropout: 0.2 | Type: BAYESIAN

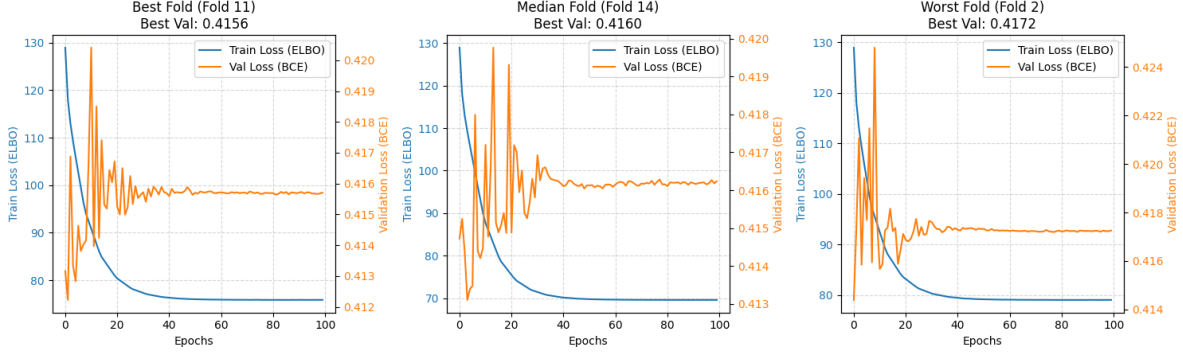


Figure 8: Training loss (ELBO, left axis) and validation loss (BCE, right axis) over 100 epochs, for the best, median and worst cross-validation fold of the Bayesian (variational) LSTM. Note the different vertical scales from Figure 2: validation BCE plateaus around 0.41–0.42, well above the 0.36–0.39 range reached by the Standard LSTM and the Deep Ensemble.

The discrimination curves in Figure 9 quantify this gap directly: across the best, median and worst folds, AUC-ROC is 0.551, 0.538 and 0.537 – barely above the 0.5 random-classifier diagonal – and AUC-PR is 0.193, 0.171 and 0.197, only marginally above the dataset’s positive-class prevalence (≈ 0.145 , the AUC-PR expected from a non-informative classifier). This is the concrete, quantitative basis for the exclusion decision stated above: under the same data, the same features and the same cross-validation protocol used for the Standard LSTM and the Deep Ensemble, the variational architecture implemented here essentially failed to learn a useful discriminative signal, despite its validation loss curve being, superficially, the most stable of the three architectures.

ROC & PR Curves | Layers: 2 | Dropout: 0.2 | Type: BAYESIAN

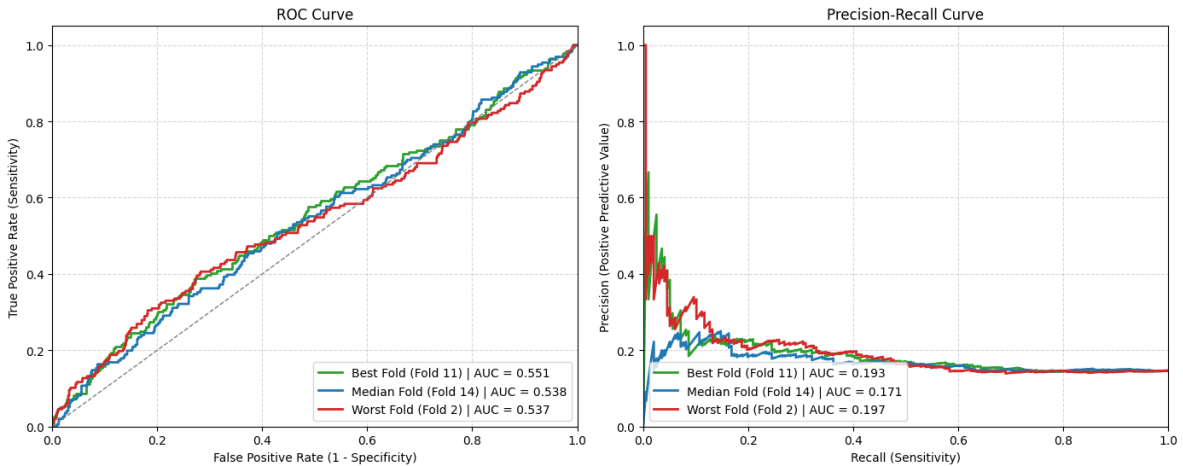


Figure 9: ROC (left) and Precision-Recall (right) curves on the validation partition for the best, median and worst cross-validation fold of the Bayesian (variational) LSTM. AUC-ROC is close to the 0.5 random-classifier diagonal and AUC-PR is close to the dataset’s positive prevalence (≈ 0.145) in all three folds, in clear contrast with Figure 3.

B Additional Discrimination and Calibration Metrics Across Missingness Scenarios

Table 3 reports the mean value, across the 15 cross-validation folds, of four additional metrics not discussed in Section 4.2 – AUC-ROC, Balanced Accuracy, the class-balanced Brier score and the Expected Calibration Error (ECE) – for the three inference strategies and all nine missingness scenarios. As with AUC-PR, the Standard LSTM and its MC-Dropout counterpart are virtually indistinguishable on every metric and every scenario (as expected, since they share the same trained weights), and the Deep Ensemble is consistently, if modestly, the best-calibrated and best-discriminating strategy – most visibly on ECE, where the Ensemble’s calibration error at 60% MCAR (0.039) remains noticeably lower than either Standard variant’s (0.044–0.045), suggesting that averaging across independently trained members may be particularly helpful for calibration under severe missingness, even where it offers only a marginal discrimination (AUC) benefit.

Table 3: Mean AUC-ROC, Balanced Accuracy, class-balanced Brier score and ECE across the 15 cross-validation folds, by inference strategy and missingness scenario.

Scenario	AUC-ROC			Balanced Accuracy			Brier Score (balanced)			ECE		
	Std	McD	Ens	Std	McD	Ens	Std	McD	Ens	Std	McD	Ens
0%	0.681	0.681	0.686	0.644	0.644	0.646	0.329	0.328	0.328	0.036	0.036	0.034
10% MCAR	0.675	0.675	0.679	0.635	0.634	0.636	0.328	0.328	0.327	0.036	0.036	0.035
20% MCAR	0.678	0.678	0.683	0.639	0.640	0.642	0.328	0.327	0.327	0.038	0.037	0.035
40% MCAR	0.668	0.668	0.672	0.634	0.634	0.635	0.329	0.328	0.328	0.038	0.038	0.034
60% MCAR	0.654	0.654	0.658	0.627	0.627	0.628	0.328	0.327	0.327	0.044	0.045	0.039
10% Temporal	0.675	0.675	0.680	0.635	0.634	0.636	0.328	0.328	0.327	0.036	0.037	0.036
20% Temporal	0.670	0.671	0.675	0.642	0.642	0.646	0.329	0.329	0.328	0.036	0.037	0.035
40% Temporal	0.648	0.648	0.651	0.624	0.624	0.626	0.330	0.330	0.329	0.038	0.039	0.036
60% Temporal	0.642	0.642	0.644	0.619	0.619	0.618	0.328	0.328	0.328	0.048	0.047	0.043

In addition to the tabular summary above, Figures 10 through 17 reproduce, for AUC-ROC, Balanced Accuracy, Recall and Precision respectively, the same pair of full-page layouts used for AUC-PR in Section 4.2: a per-scenario bar chart, in the style of Figure 4, showing the mean value ($\pm 95\%$ CI) of the three inference strategies across all nine missingness scenarios, together with the corresponding omnibus (repeated-measures ANOVA and Friedman) tests and, where significant, the Holm-corrected pairwise Wilcoxon signed-rank post-hoc comparisons; and a forest plot, in the style of Figure 5, reorganizing the same values as a function of missingness severity (%NA), separately for each inference strategy and for the MCAR and Temporal-Block mechanisms. This extends the same scenario-by-scenario and severity-by-severity statistical comparison performed for AUC-PR to each of these four complementary discrimination and classification metrics.

AUC-ROC | Paired Statistical Analysis Grid



Figure 10: AUC-ROC (mean $\pm$ 95% CI) for the three inference strategies (standard, MC-Dropout, ensemble; $n = 15$ folds each) across all nine missingness scenarios (left column), together with omnibus repeated-measures ANOVA and Friedman tests and, where significant, Holm-corrected pairwise Wilcoxon signed-rank post-hoc comparisons (right column).

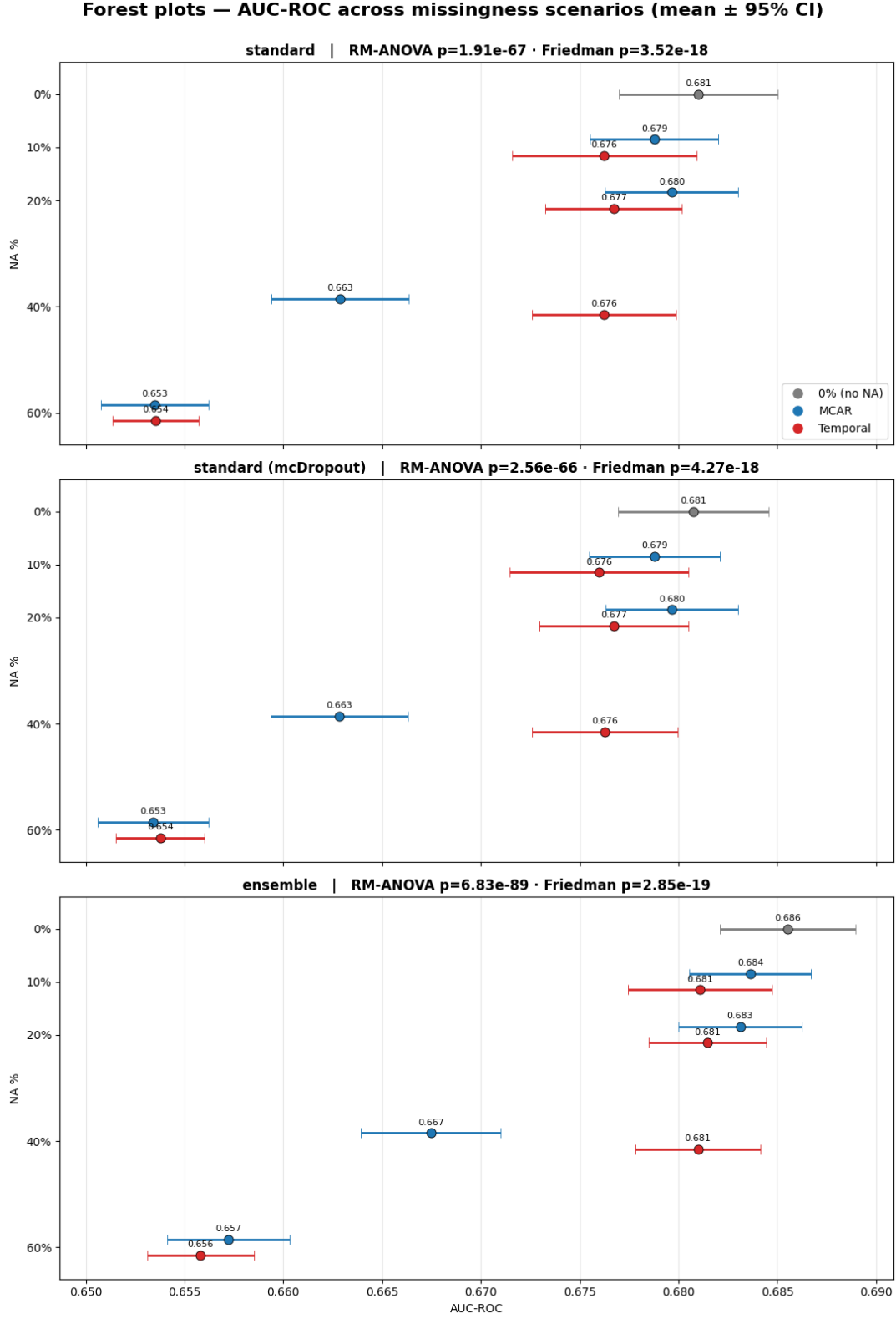


Figure 11: Forest plot of mean AUC-ROC ($\pm 95\%$ CI) as a function of missingness severity (%NA), separately for the Standard LSTM (top), MC-Dropout (middle) and Deep Ensemble (bottom), and for the MCAR (blue) and Temporal-Block (red) mechanisms. The grey 0% point is the shared baseline. RM-ANOVA and Friedman p -values across the nine scenarios are annotated per panel.

Balanced_Accuracy | Paired Statistical Analysis Grid

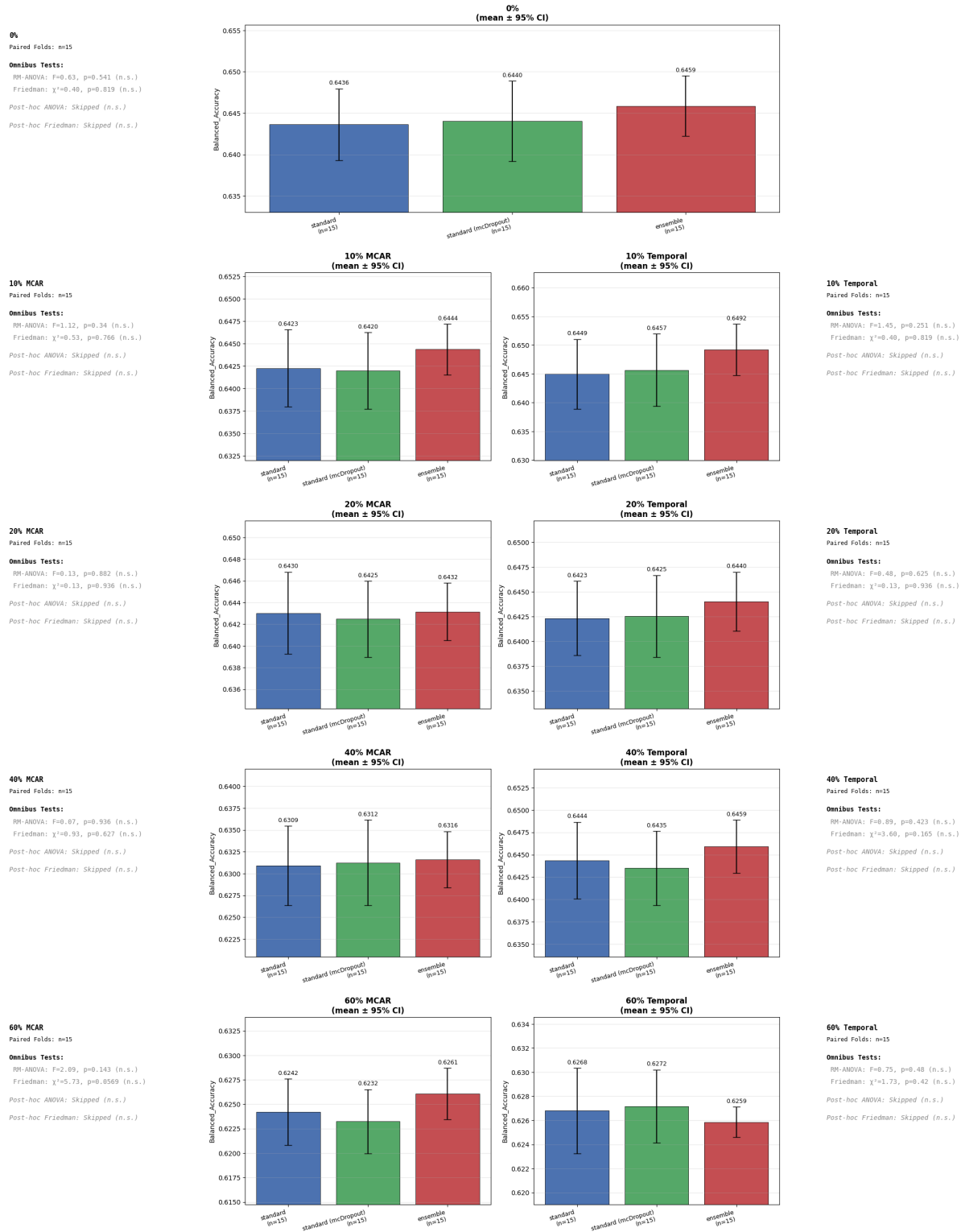


Figure 12: Balanced Accuracy (mean $\pm$ 95% CI) for the three inference strategies (standard, MC-Dropout, ensemble; $n = 15$ folds each) across all nine missingness scenarios (left column), together with omnibus repeated-measures ANOVA and Friedman tests and, where significant, Holm-corrected pairwise Wilcoxon signed-rank post-hoc comparisons (right column).

Forest plots — Balanced_Accuracy across missingness scenarios (mean $\pm$ 95% CI)

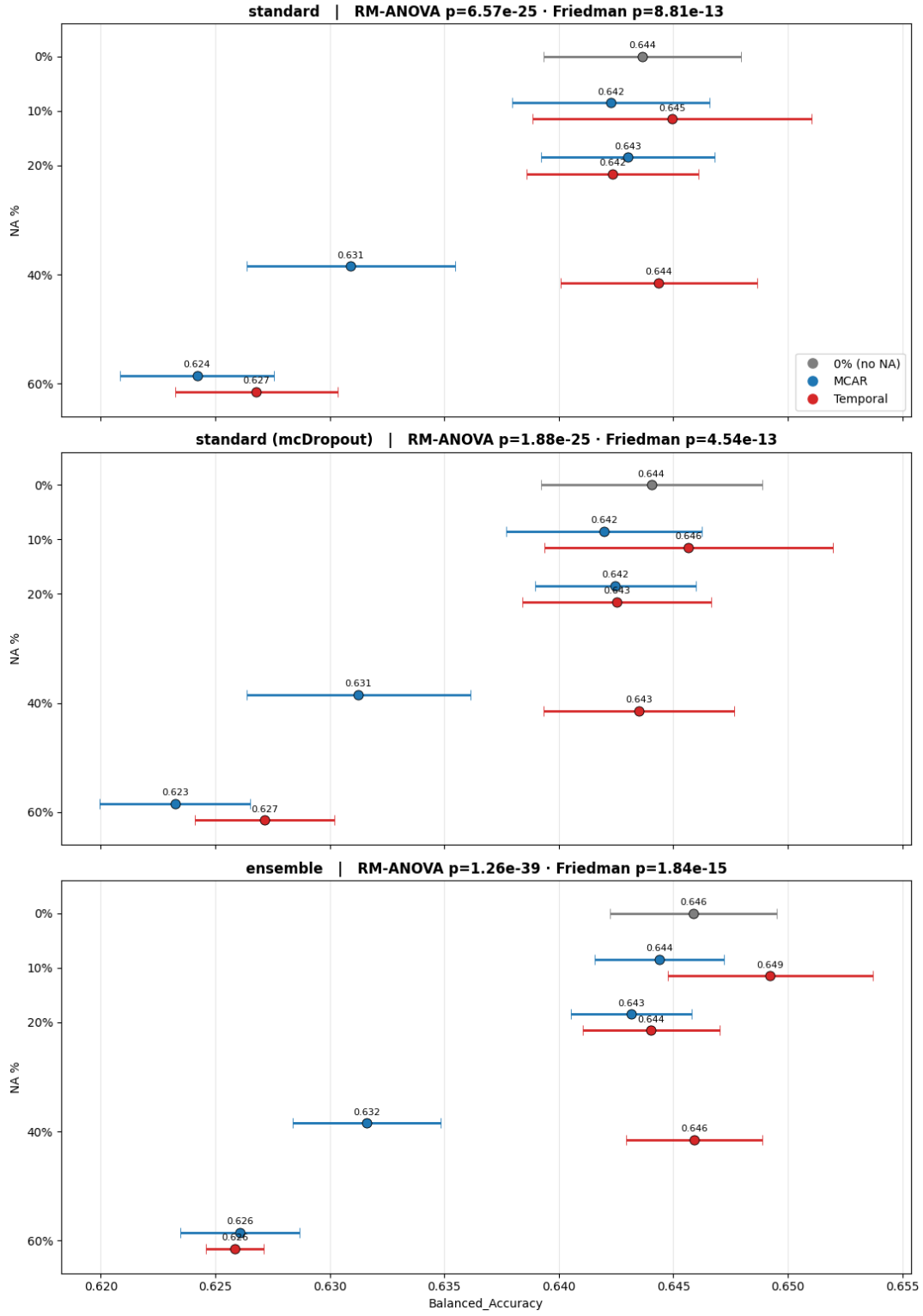


Figure 13: Forest plot of mean Balanced Accuracy ($\pm$ 95% CI) as a function of missingness severity (%NA), separately for the Standard LSTM (top), MC-Dropout (middle) and Deep Ensemble (bottom), and for the MCAR (blue) and Temporal-Block (red) mechanisms. The grey 0% point is the shared baseline. RM-ANOVA and Friedman p -values across the nine scenarios are annotated per panel.

Recall (Sens) | Paired Statistical Analysis Grid

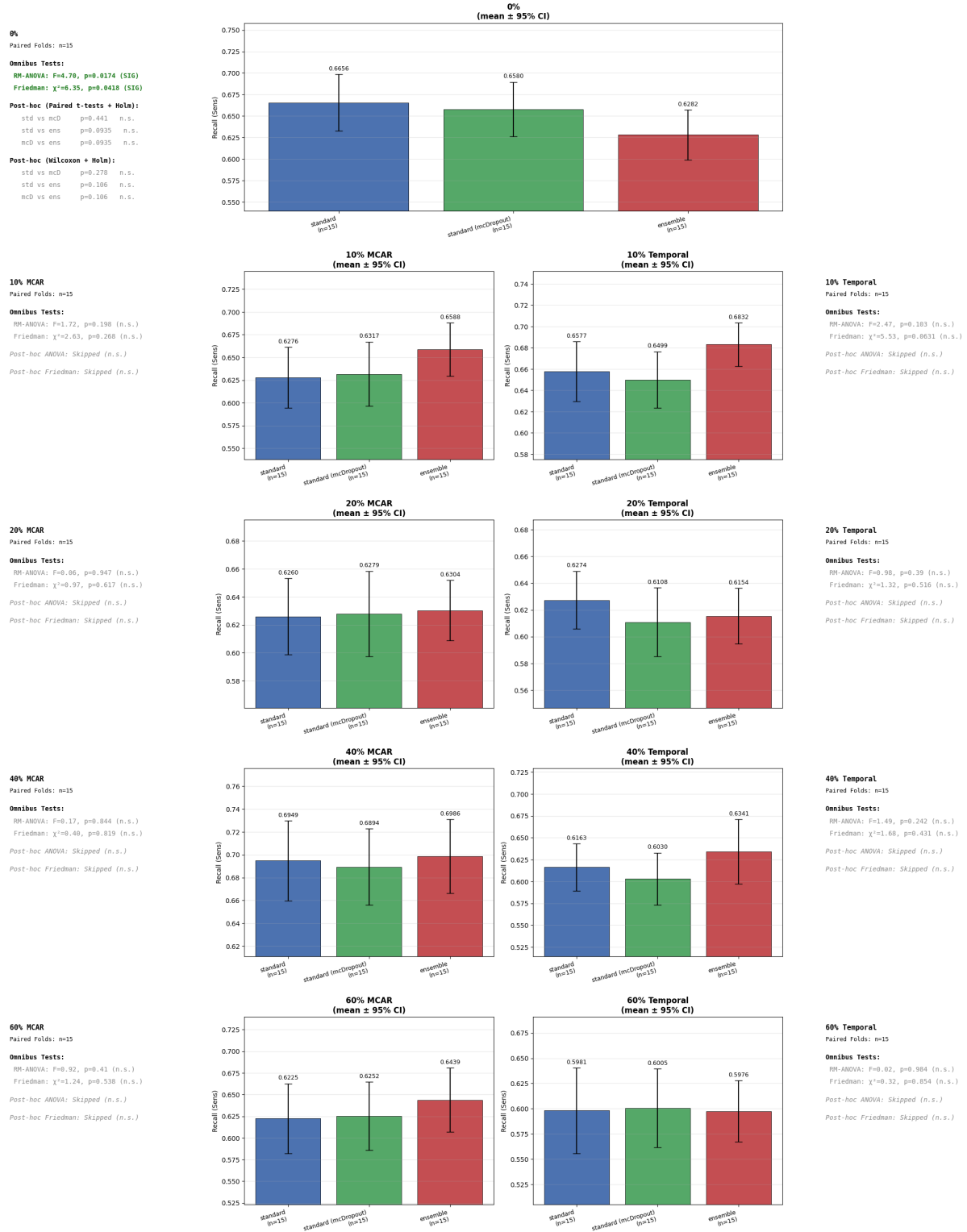


Figure 14: Recall (mean $\pm$ 95% CI) for the three inference strategies (standard, MC-Dropout, ensemble; $n = 15$ folds each) across all nine missingness scenarios (left column), together with omnibus repeated-measures ANOVA and Friedman tests and, where significant, Holm-corrected pairwise Wilcoxon signed-rank post-hoc comparisons (right column).

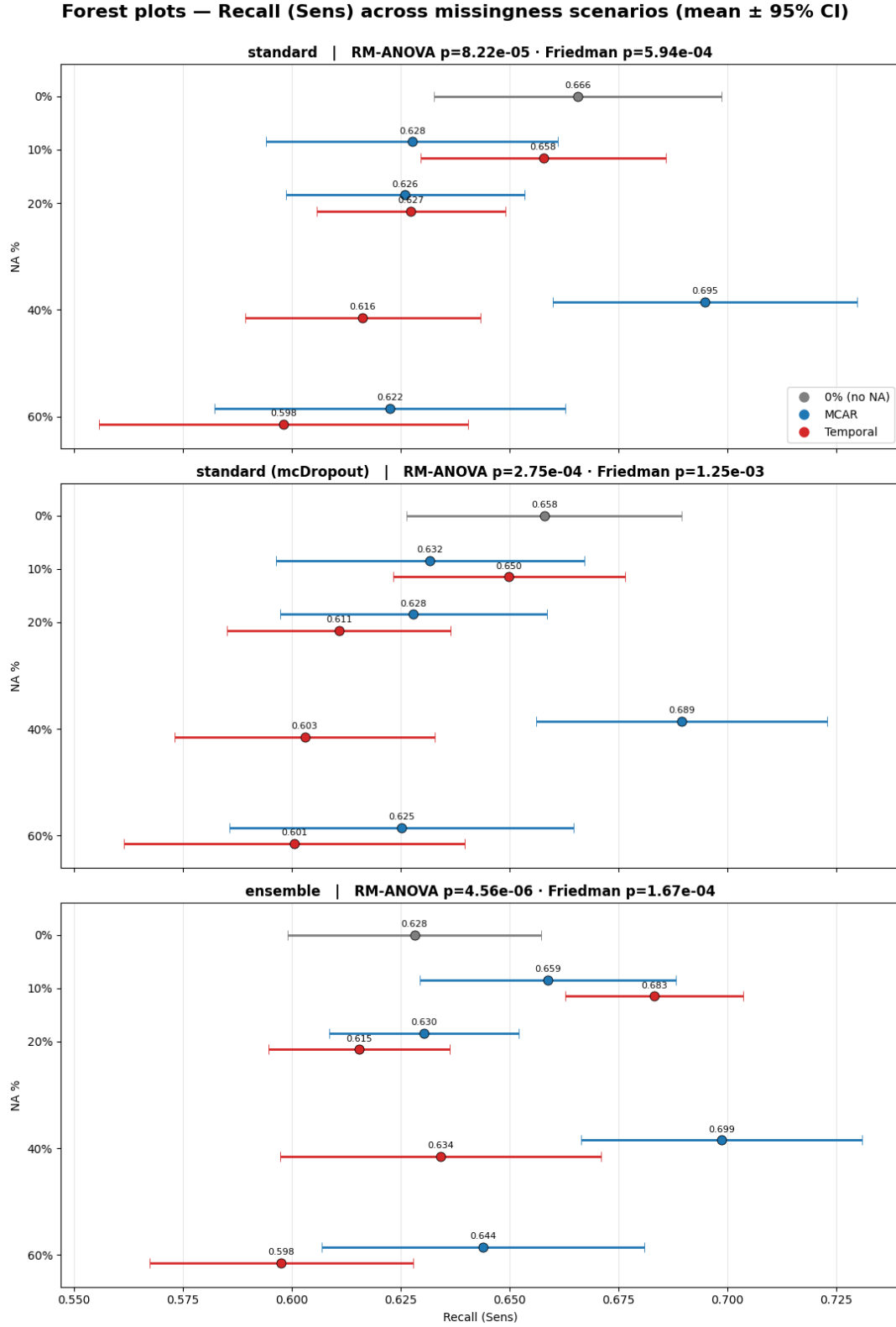


Figure 15: Forest plot of mean Recall ($\pm 95\%$ CI) as a function of missingness severity (%NA), separately for the Standard LSTM (top), MC-Dropout (middle) and Deep Ensemble (bottom), and for the MCAR (blue) and Temporal-Block (red) mechanisms. The grey 0% point is the shared baseline. RM-ANOVA and Friedman p -values across the nine scenarios are annotated per panel.

Precision (PPV) | Paired Statistical Analysis Grid

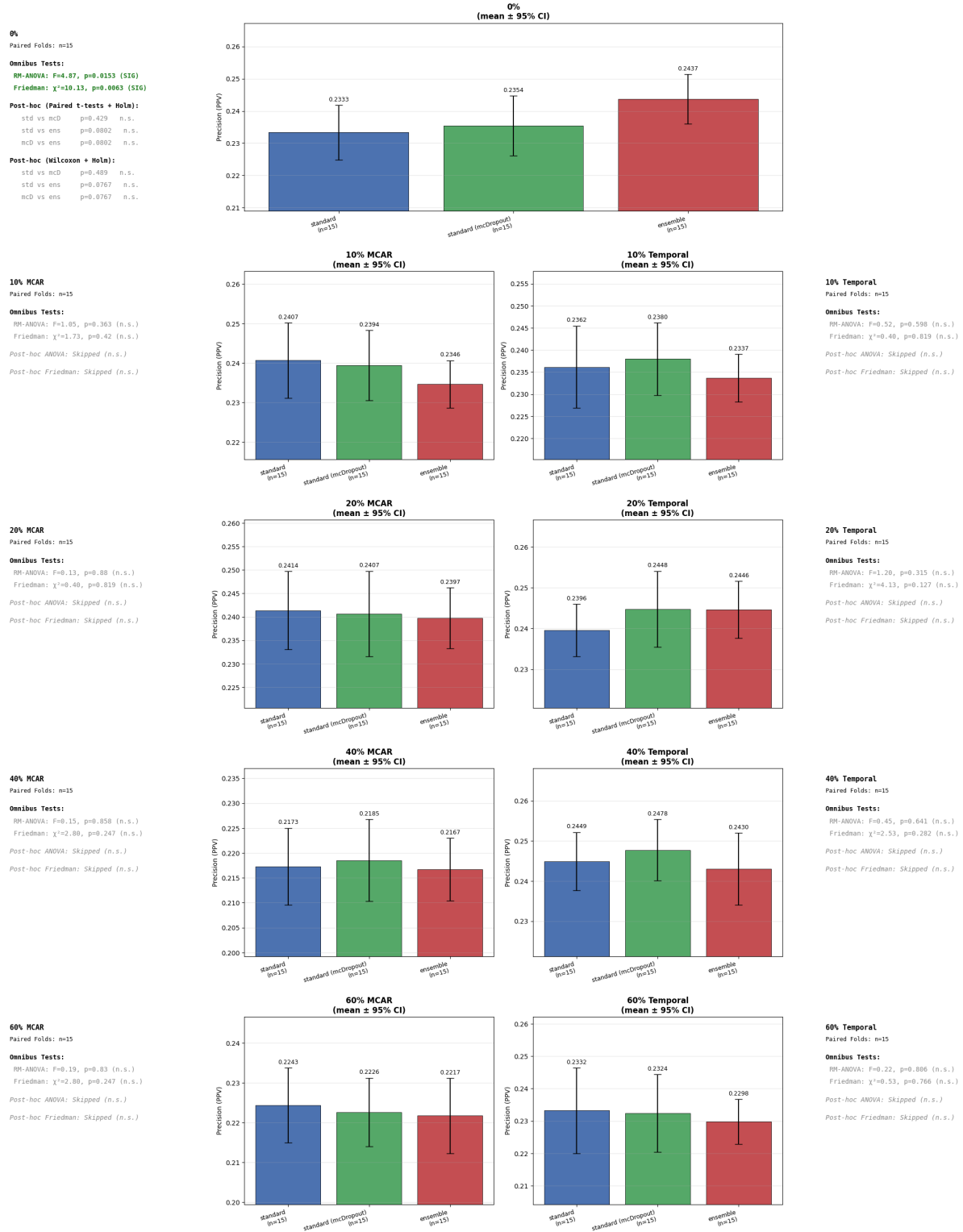


Figure 16: Precision (mean $\pm$ 95% CI) for the three inference strategies (standard, MC-Dropout, ensemble; $n = 15$ folds each) across all nine missingness scenarios (left column), together with omnibus repeated-measures ANOVA and Friedman tests and, where significant, Holm-corrected pairwise Wilcoxon signed-rank post-hoc comparisons (right column).

C Missing-Data Statistics Across the Nine Test Scenarios

Table 4 reports the per-feature percentage of missing values for the test set under the 0% baseline and under each of the eight injected scenarios. Because the injected percentage is computed relative to each patient’s *currently available* observations rather than to a fixed denominator (Section 3.1.10), MCAR and Temporal-Block injection at the same nominal severity always yield identical aggregate per-feature missingness statistics; the two mechanisms differ only in *which* time steps are removed, not in *how many*. The last two columns confirm two facts stated in Section 4.2: GCS and Temperature already exceed 50% missing at the 0% baseline, and the overall mean missingness per feature rises from 26.5% at baseline to 70.7% at the 60% injection level.

Table 4: Per-feature percentage of missing values (%) in the test set, before imputation, for the 0% baseline and each injected scenario. MCAR and Temporal scenarios at the same severity share identical aggregate statistics by construction (see text).

Scenario	GCS	HR	Temp	Urine	DiasABPf	MAPf	SysABPf	Mean	Std
0% (baseline)	60.71	9.94	56.39	26.26	10.55	11.10	10.54	26.50	25.67
10% MCAR / Temporal	69.04	18.27	64.73	34.60	18.88	19.44	18.87	34.83	25.67
20% MCAR / Temporal	70.14	26.40	66.47	40.76	26.99	27.53	26.98	40.75	22.08
40% MCAR / Temporal	76.58	45.76	73.84	55.68	46.10	46.46	46.10	55.79	15.62
60% MCAR / Temporal	84.13	64.17	82.56	70.59	64.45	64.64	64.44	70.71	10.22

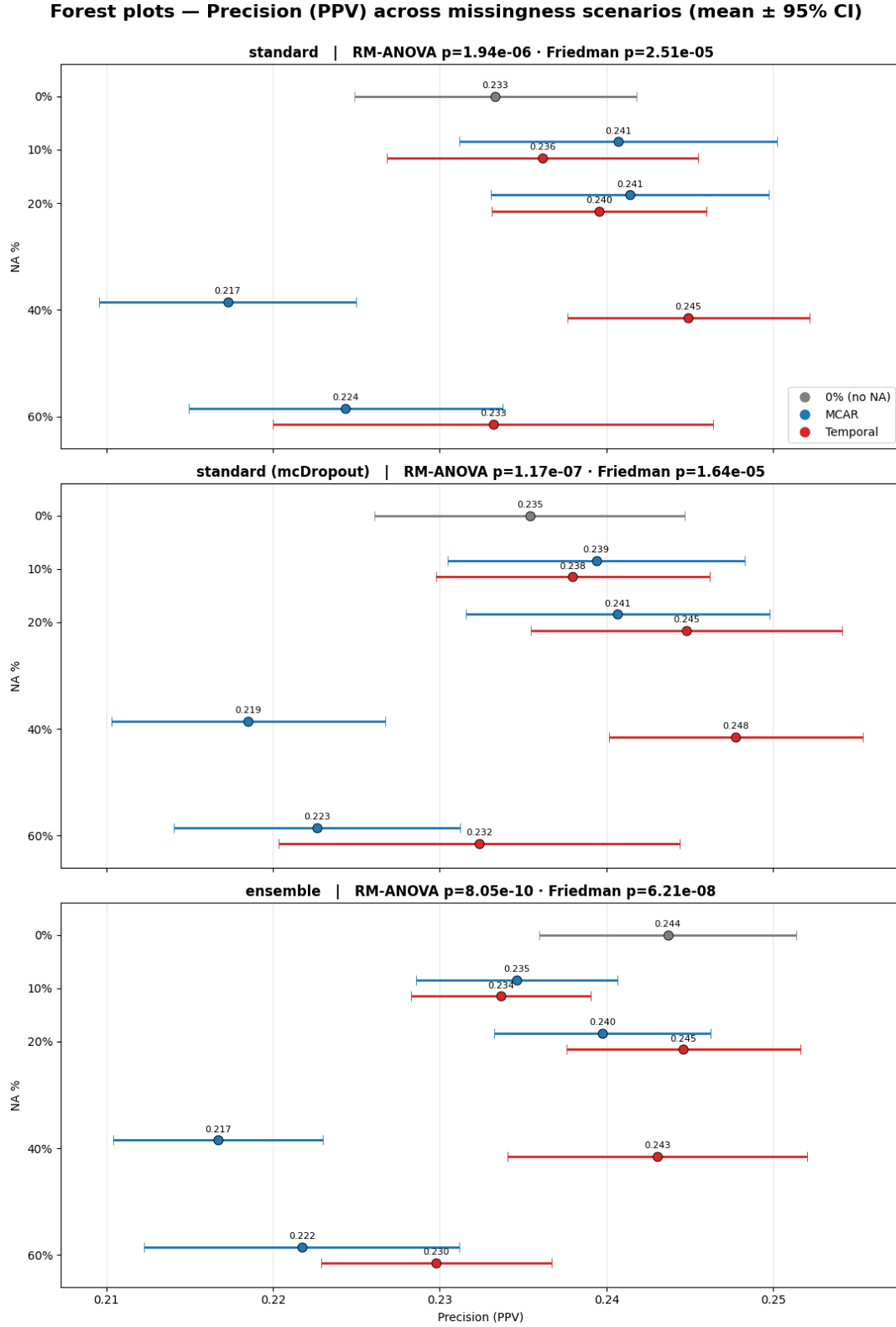


Figure 17: Forest plot of mean Precision ($\pm 95\%$ CI) as a function of missingness severity (%NA), separately for the Standard LSTM (top), MC-Dropout (middle) and Deep Ensemble (bottom), and for the MCAR (blue) and Temporal-Block (red) mechanisms. The grey 0% point is the shared baseline. RM-ANOVA and Friedman p -values across the nine scenarios are annotated per panel.

D Patient Counts by Error-Agreement Category

Table 5 reports the table of number of categorized predictions by model: the absolute number of test-set patients falling into each of the four original error-agreement categories (Confident Correct, Confident Error, Borderline Correct, Borderline Error; see Section 4.3), for each of the three representative predictors (Standard fold 2, MC-Dropout fold 2, Ensemble fold 9) and each of the nine missingness scenarios, with the MCAR and Temporal-Block mechanisms listed as separate rows at every severity level. The corresponding trends are discussed visually in Figure 6.

Table 5: Number of categorized predictions by model: patient counts per error-agreement category, predictor and missingness scenario.

Category	Scenario	Standard	MC-Dropout	Ensemble
Confident Correct	0%	515	502	539
	10% MCAR	502	476	527
	10% Temporal	502	488	527
	20% MCAR	506	497	526
	20% Temporal	494	484	527
	40% MCAR	492	483	520
	40% Temporal	465	458	502
	60% MCAR	462	459	483
	60% Temporal	465	460	482
Confident Error	0%	424	425	473
	10% MCAR	460	460	491
	10% Temporal	450	456	488
	20% MCAR	451	454	490
	20% Temporal	451	453	489
	40% MCAR	470	474	498
	40% Temporal	484	483	557
	60% MCAR	506	515	563
	60% Temporal	536	536	587
Borderline Correct	0%	441	443	376
	10% MCAR	430	449	378
	10% Temporal	427	440	371
	20% MCAR	424	421	363
	20% Temporal	429	438	381
	40% MCAR	422	416	359
	40% Temporal	419	415	336
	60% MCAR	405	407	342
	60% Temporal	398	389	333
Borderline Error	0%	304	314	296
	10% MCAR	292	299	288
	10% Temporal	305	300	298
	20% MCAR	303	312	305
	20% Temporal	310	309	287
	40% MCAR	300	311	307
	40% Temporal	316	328	289
	60% MCAR	311	303	296
	60% Temporal	285	299	282

References

- [1] Z. C. Lipton, D. C. Kale, and R. Wetzel. Directly modeling missing data in sequences with RNNs: Improved classification of clinical time series. In *Proceedings of the 1st Machine Learning for Healthcare Conference*, PMLR 56, pages 253–270, 2016.
- [2] Z. Che, S. Purushotham, K. Cho, D. Sontag, and Y. Liu. Recurrent neural networks for multivariate time series with missing values. *Scientific Reports*, 8:6085, 2018.
- [3] A survey of using EHR as real-world evidence for discovering and validating new drug indications. arXiv:2505.24767, 2025.
- [4] H. C. Thorsen-Meyer et al. Dynamic and explainable machine learning prediction of mortality in patients in the intensive care unit: a retrospective study of high-frequency data in electronic patient records. *The Lancet Digital Health*, 2(4):e179–e191, 2020.
- [5] H. Harutyunyan, H. Khachatrian, D. C. Kale, G. Ver Steeg, and A. Galstyan. Multitask learning and benchmarking with clinical time series data. *Scientific Data*, 6:96, 2019.
- [6] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed., ch. 7, pp. 241–249. Springer Science & Business Media, New York, NY, USA, 2009.
- [7] Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning*, PMLR 48, pages 1050–1059, 2016.
- [8] B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017.
- [9] J. Xia, J. Zhang, S. Zhang, C. Liu, Z. Liu, Y. Zhao, H. Zhang, and M. Shen. Development and validation of an explainable machine learning model for mortality prediction in ICU patients with lung cancer. *Frontiers in Cell and Developmental Biology*, 14:1862263, 2026. doi: 10.3389/fcell.2026.1862263
- [10] Inadequacy of common stochastic neural networks for reliable clinical decision support. arXiv:2401.13657, 2024.
- [11] I. Silva, G. Moody, D. J. Scott, L. A. Celi, and R. G. Mark. Predicting in-hospital mortality of ICU patients: The PhysioNet/Computing in Cardiology Challenge 2012. *Computing in Cardiology*, 39:245–248, 2012.
- [12] M. Kim and S.-Y. Shin. Temporal Dynamic Embedding for Irregularly Sampled Time Series. arXiv:2504.05768, 2025.
- [13] A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola. *Dive into Deep Learning*, Chapter 10.1: Long Short-Term Memory (LSTM). Cambridge University Press, 2023. Available at: https://d2l.ai/chapter_recurrent-modern/lstm.html
- [14] T. Saito and M. Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432, 2015.

- [15] J. Davis and M. Goadrich. The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning (ICML 2006)*, pages 233–240, 2006.
- [16] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.
- [17] T. Papamarkou, M. Skoularidou, K. Palla, L. Aitchison, J. Arbel, D. Dunson, M. Filippone, V. Fortuin, P. Hennig, J. M. Hernández-Lobato, A. Hubin, A. Immer, T. Karaletsos, M. E. Khan, A. Kristiadi, Y. Li, S. Mandt, C. Nemeth, M. A. Osborne, T. G. J. Rudner, D. Rügamer, Y. W. Teh, M. Welling, A. G. Wilson, and R. Zhang. Position: Bayesian deep learning is needed in the age of large-scale AI. In *Proceedings of the 41st International Conference on Machine Learning*, PMLR 235, pages 39556–39586, 2024.